

1. Gearing up and down
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 - i. Single-Speed Spur Gearboxes
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 - iii. Worm Gearboxes
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Quick Comparison

Gearbox Type	Best For	Main Advantage	Main Concern
Single-Speed Spur	Drivetrains, elevators, climbers, simple mechanisms	Simple and reliable	Only one speed/torque compromise
Dual-Speed Shifting	Tank drivetrains, speed + pushing games	Switches between speed and torque	More complexity and maintenance
Worm	Arms, wrists, rotating mechanisms	High reduction and back-drive resistance	Lower efficiency and heat
Right-Angle	Swerve, tight packaging, sideways motor placement	Changes power direction	Alignment and efficiency
Planetary	Arms, intakes, wrists, climbers, prototypes	Compact modular reduction	Side loads and shock loads

Simple FRC Design Rule

Use the simplest gearbox that meets the need:

Use single-speed spur when you need a strong, simple fixed ratio.

Use dual-speed shifting only when the strategy truly benefits from both high speed and low-speed torque.

Use worm gearboxes for compact, slow, high-reduction rotation where efficiency is less important.

Use right-angle gearboxes when packaging requires the motor and output shaft to be perpendicular.

Use **planetary gearboxes** when you need compact, modular reduction for a mechanism.

Single-Speed Spur Gearboxes

A **single-speed spur gearbox** uses spur gears to create one fixed gear ratio. Spur gears are the common flat gears with teeth around the outside. The input gear turns another gear on a parallel shaft, changing motor speed and torque.

How it works	• Small gear driving large gear = gear reduction. • Gear reduction lowers output speed but increases output torque. • The gearbox has one fixed ratio, so the robot cannot shift between “fast” and “torque” modes during a match. • These are common because they are simple, strong, and easy to repair.
FRC examples	• Drivetrain gearboxes for tank drive or West Coast Drive. • Single-stage or two-stage drive gearboxes from West Coast Products. • AndyMark ToughBox-style drivetrain gearboxes. • Custom gearboxes for elevators, climbers, or arms. • REV’s drivetrain gearboxes are also examples of compact FRC gearboxes used to transform motor motion. REV describes gearboxes as common, compact, modular ways to transform motion in FRC.
Pros	• Simple to design, build, and troubleshoot. • Reliable and durable. • Easier for new students to understand. • Fewer moving parts than shifting gearboxes. • Usually lighter and cheaper than dual-speed designs. • Good for modern FRC drivetrains, especially with powerful brushless motors.
Cons	• One ratio must compromise between speed and pushing force. • If geared too fast, the robot may accelerate poorly or draw too much current. • If geared too slow, the robot may be easy to control but lose cycle time. • Cannot switch into a lower gear for pushing or climbing torque.
When to use in FRC	• Most drivetrains. • Simple, reliable competition robots. • Teams prioritizing robustness over complexity.

	• Mechanisms that only need one speed range.
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Dual-Speed Shifting Gearboxes

A **dual-speed shifting gearbox** lets the robot switch between two gear ratios, usually a **high gear** for speed and a **low gear** for torque.

How it works	• The gearbox contains two possible gear paths. • A dog gear, shifting dog, or similar mechanism locks one ratio at a time • Shifting may be pneumatic, servo-driven, or mechanically actuated • High gear gives more speed but less torque. • Low gear gives more torque but less speed.
FRC examples	• AndyMark EVO Shifter. • WCP dog-shifting drive gearboxes. • Older AM Shifter, Super Shifter, or Sonic Shifter style transmissions. • High-speed tank drive robots that want sprint speed and pushing power. • Some specialized mechanisms that need both fast travel and strong final positioning.
Pros	• High gear can improve full-field driving speed. • Low gear can reduce current draw during pushing or acceleration. • Useful for games with long driving distances plus heavy defense. • Gives drivers more options. • Can help avoid breaker trips when pushing in low gear.
Cons	• More complex mechanically. • More parts to fail. • Requires shifting hardware and software logic. • Heavier than single-speed gearboxes. • Takes more space. • Less necessary now that modern brushless motors are powerful and controllable. • Drivers must know when to shift, or code must handle it automatically.

When to use in FRC	• Tank-drive robots that need both sprint speed and pushing power. • Games with long open-field cycles and heavy defense. • Teams with experience maintaining shifting drivetrains. • Less common for teams using swerve because swerve modules already add drivetrain complexity
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Worm Gearboxes

A **worm gearbox** uses a screw-like worm gear to drive a gear wheel. The input and output shafts are usually at 90 degrees to each other.

How it works	• The worm looks like a threaded screw. • The worm turns the worm wheel. • This creates a large reduction in a compact space. • Worm gearboxes often resist being back-driven, though teams should not treat that as a guaranteed safety brake.
FRC examples	• AndyMark RAW Box. • Arms or wrists that need high reduction. • Rotating mechanisms that benefit from holding position. • Compact right-angle mechanisms. • Some older turret, hood, or manipulator designs.
Pros	• High gear reduction in a compact package. • Turns power 90 degrees. • Can resist back-driving better than many spur gear reductions. • Useful for arms, wrists, and rotating mechanisms. • Can simplify packaging when the motor cannot be inline with the output.
Cons	• Less efficient than spur or planetary gearboxes. • Can generate more heat due to sliding friction. • Needs lubrication. • Can wear faster if overloaded or poorly supported. • Usually not ideal for high-speed continuous motion. • Back-drive resistance should not be relied on as the only way to hold a heavy or dangerous mechanism.

When to use in FRC	• Low-speed, high-torque mechanisms. • Wrists, hoods, turrets, and arms. • Situations where right-angle packaging is useful. • Mechanisms where some resistance to back-driving is helpful.
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Right-Angle Gearboxes

A **right-angle gearbox** changes the direction of rotation by 90 degrees. This can be done with bevel gears, miter gears, or worm gears.

How it works	• The input shaft and output shaft are perpendicular. • Bevel gears transfer power between angled shafts. • Worm gearboxes also create right-angle power transfer, but with more reduction and more friction. • Right-angle gearboxes are usually chosen because of packaging, not just ratio.
FRC examples	• Swerve modules, where bevel gears transfer motor power down to the wheel. • AndyMark 0.375 in. hex pass-through bevel box. • Worm-based right-angle boxes like the RAW Box. • Intake or conveyor systems where the motor must be mounted sideways. • Compact arms or wrists where motor placement is limited.
Pros	• Lets the motor fit in a more convenient location. • Useful when space is tight. • Common in swerve modules. • Can simplify chain, belt, or shaft routing. • Can protect motors by moving them away from impact zones.
Cons	• More alignment-sensitive than simple spur reductions. • Bevel gears can be noisy or wear if poorly aligned. • May be less efficient than simple spur gears. • Can be harder to inspect and repair. • Often requires careful bearing support.

When to use in FRC	• Swerve drive modules. • Tight packaging situations. • Mechanisms where the motor shaft direction does not match the desired output shaft direction. • Designs where moving the motor improves serviceability or protection.
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Planetary Gearboxes

A **planetary gearbox** uses a central sun gear, planet gears, and an outer ring gear. The planets rotate around the sun gear, creating a compact reduction.

How it works	• The motor drives the sun gear. • Planet gears share the load. • The output usually comes from a carrier connected to the planet gears. • Multiple stages can be stacked to create larger reductions.
FRC examples	• REV MAXPlanetary. • REV UltraPlanetary. • AndyMark Sport Gearbox. • AndyMark PG gearboxes. • Arm, wrist, intake, conveyor, shooter-adjustment, and climber mechanisms. • Prototyping mechanisms where teams want to swap ratios quickly.
Pros	• Compact for the amount of reduction. • Easy to mount to many FRC motors. • Good torque density. • Modular ratio options. • Great for prototyping. • Often easier than designing a custom gearbox. • Useful for arms, intakes, wrists, climbers, and rollers.
Cons	• Can fail if overloaded or shock-loaded. • Some planetary gearboxes do not handle large side loads well.

	• Output shafts may need external support. • Multiple stacked stages can add friction and backlash. • Not always ideal for drivetrains unless specifically designed for that purpose. • Teams must check torque limits and allowed motor combinations.
When to use in FRC	• Mechanisms needing compact reduction. • Prototypes where ratio changes are likely. • Intakes, indexers, arms, wrists, elevators, climbers, and light-to-medium duty mechanisms. • When a COTS solution saves design and manufacturing time.